Multimodal integration reveals complementary imaging and genetic predictors of coronary artery disease risk

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github.com/Devanshpandey/Multimodal-integration-for-coronary-artery-disease-risk-prediction

Cardiac MRI embeddings and polygenic risk capture independent CAD risk — combining them identifies a subgroup with ≈23× the cohort event rate.

Colour key 🟧 imaging (cardiac MRI) 🟧 extra-cardiac imaging 🟦 genetics (PRS) ⬜ clinical / guideline scores ⬛ combined model

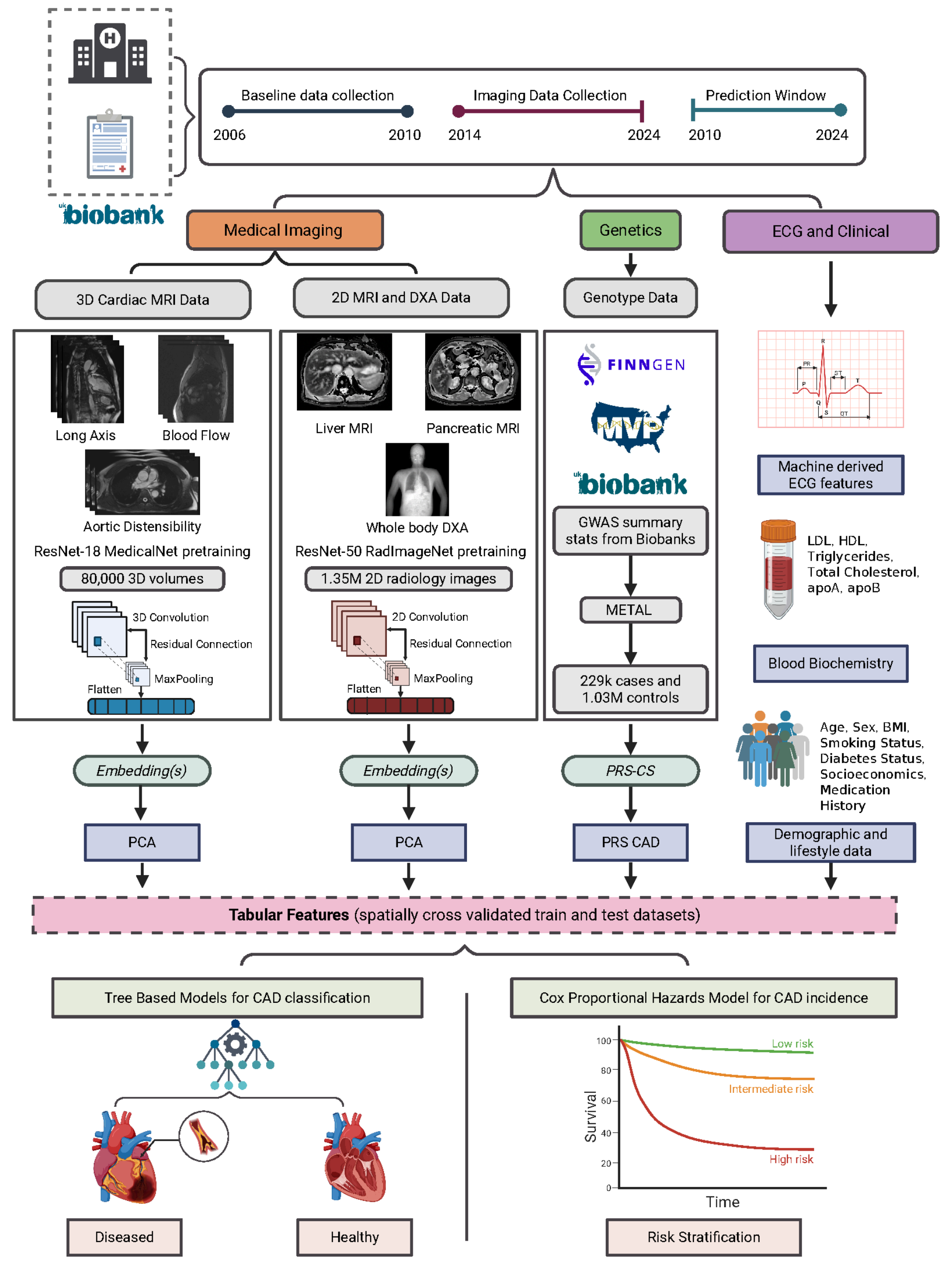
WHY & HOW

UK Biobank imaging cohort, three data domains, one model

Background & questions

- Leading cause of death; ~50% of CAD first presents as a heart attack.
- Clinical scores omit genetics and imaging; how much imaging and genetic risk overlap was unknown.

1. Which imaging modalities add to clinical, metabolic, ECG and genetic factors?
2. Overlap between imaging-derived and polygenic risk
3. Sequential integration: discrimination, calibration, stratification


biobank
Baseline data collection
2006
2010
Imaging Data Collection
2014
2024
Prediction Window
2010
2024
Medical Imaging
Genetics
ECG and Clinical
3D Cardiac MRI Data
2D MRI and DXA Data
Genotype Data
Long Axis
Blood Flow
Aortic Distensibility
ResNet-18 MedicalNet pretraining
80,000 3D volumes
3D Convolution
Residual Connection
MaxPooling
Flatten
Liver MRI
Pancreatic MRI
Whole body DXA
ResNet-50 RadImageNet pretraining
1.35M 2D radiology images
2D Convolution
Residual Connection
MaxPooling
Flatten
FINNGEN
MVP
biobank
GWAS summary stats from Biobanks
METAL
229k cases and 1.03M controls
Machine derived ECG features
LDL, HDL, Triglycerides, Total Cholesterol, apoA, apoB
Blood Biochemistry
Age, Sex, BMI, Smoking Status, Diabetes Status, Socioeconomics, Medication History
Embedding(s)
Embedding(s)
PRS-CS
PCA
PCA
PRS CAD
Demographic and lifestyle data
Tabular Features (spatially cross validated train and test datasets)
Tree Based Models for CAD classification
Cox Proportional Hazards Model for CAD incidence
Diseased
Healthy
Survival
Time
Low risk
Intermediate risk
High risk
Risk Stratification

42,351
UK Biobank imaging participants

677
incident CAD (1.6%)

4.6 yr
median follow-up

211,755
cardiac cine MRI series

127,053
2D images (liver, pancreas, DXA)

10,597
held-out test set (178 cases)

Cardiac MRI does not image coronary atherosclerosis; its signal reflects remodeling, vascular stiffness and cardiometabolic burden.

Outcome = clinically recognized CAD (MI, angina, revascularization); prevalent cases excluded. Fig. 1 of the manuscript: three data domains → tabular features → classification and Cox survival models.

Methods in brief

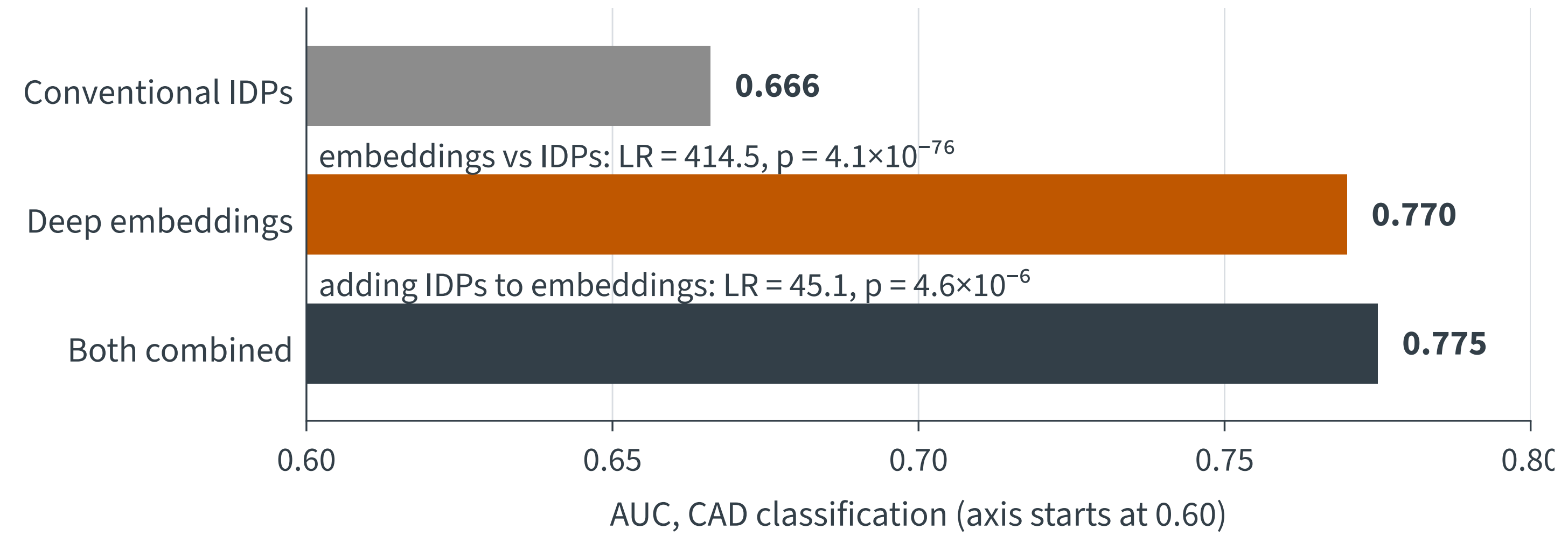
- Encoders fine-tuned on balanced case–control training data only.
- Embeddings → PCA, 10 PCs per stream (80 total).
- PRS-CS, European meta-GWAS: OR 1.87 per SD, AUC 0.766 (with age, sex, 20 PCs).
- XGBoost + nested Cox; imaging risk score = cross-fitted ridge Cox.

IMAGING REPRESENTATIONS

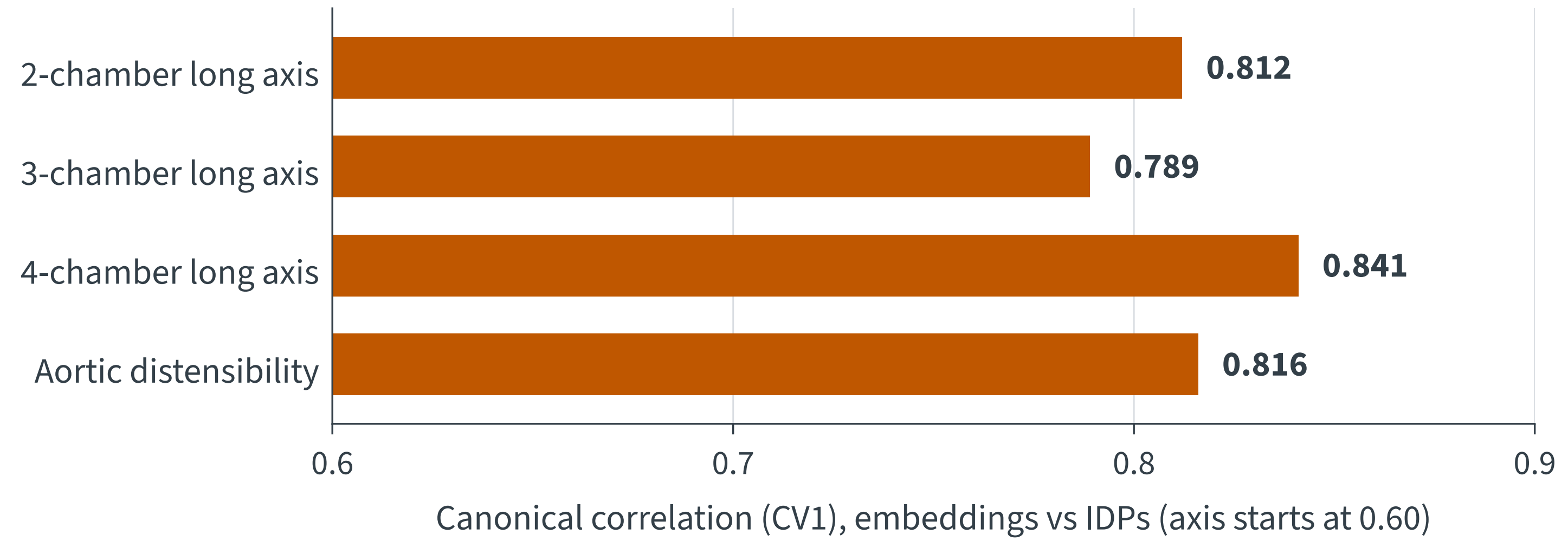
Deep embeddings beat handcrafted cardiac phenotypes

0.770 vs 0.666

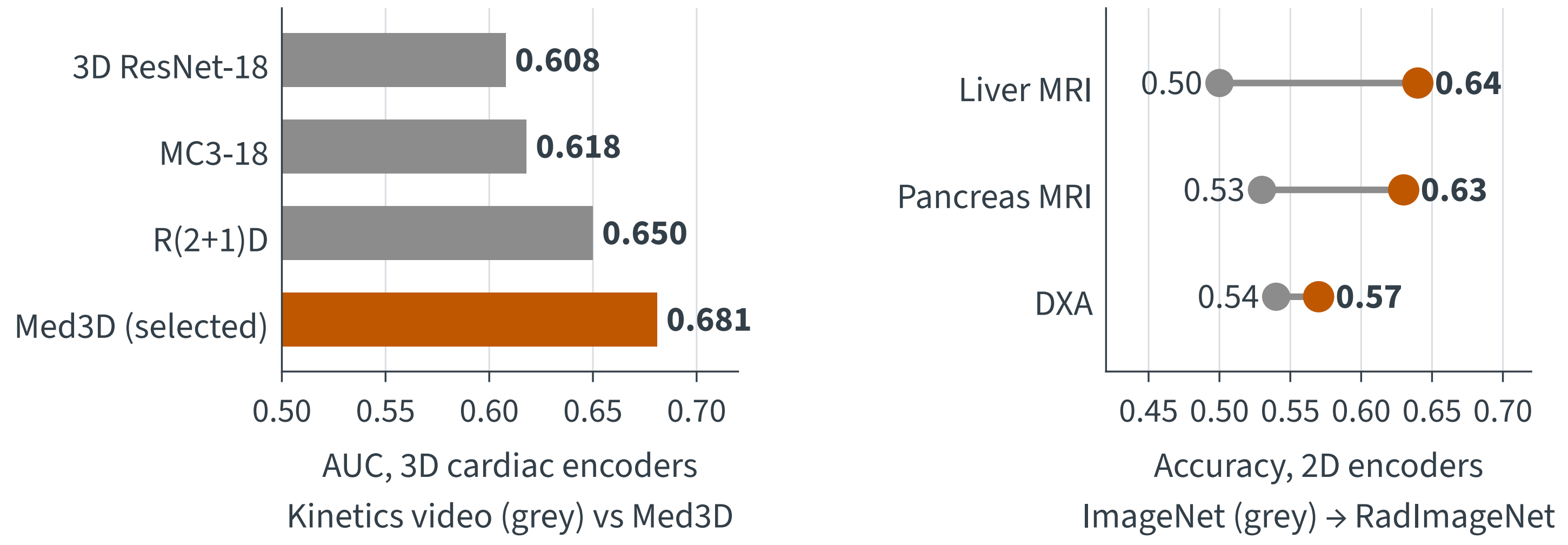
AUC, deep embeddings vs conventional imaging-derived phenotypes (IDPs); both combined 0.775



Embeddings outperform IDPs; adding IDPs barely helps (logistic models; performance plateaus at ~5 PCs per modality; Fig. 2a,b).



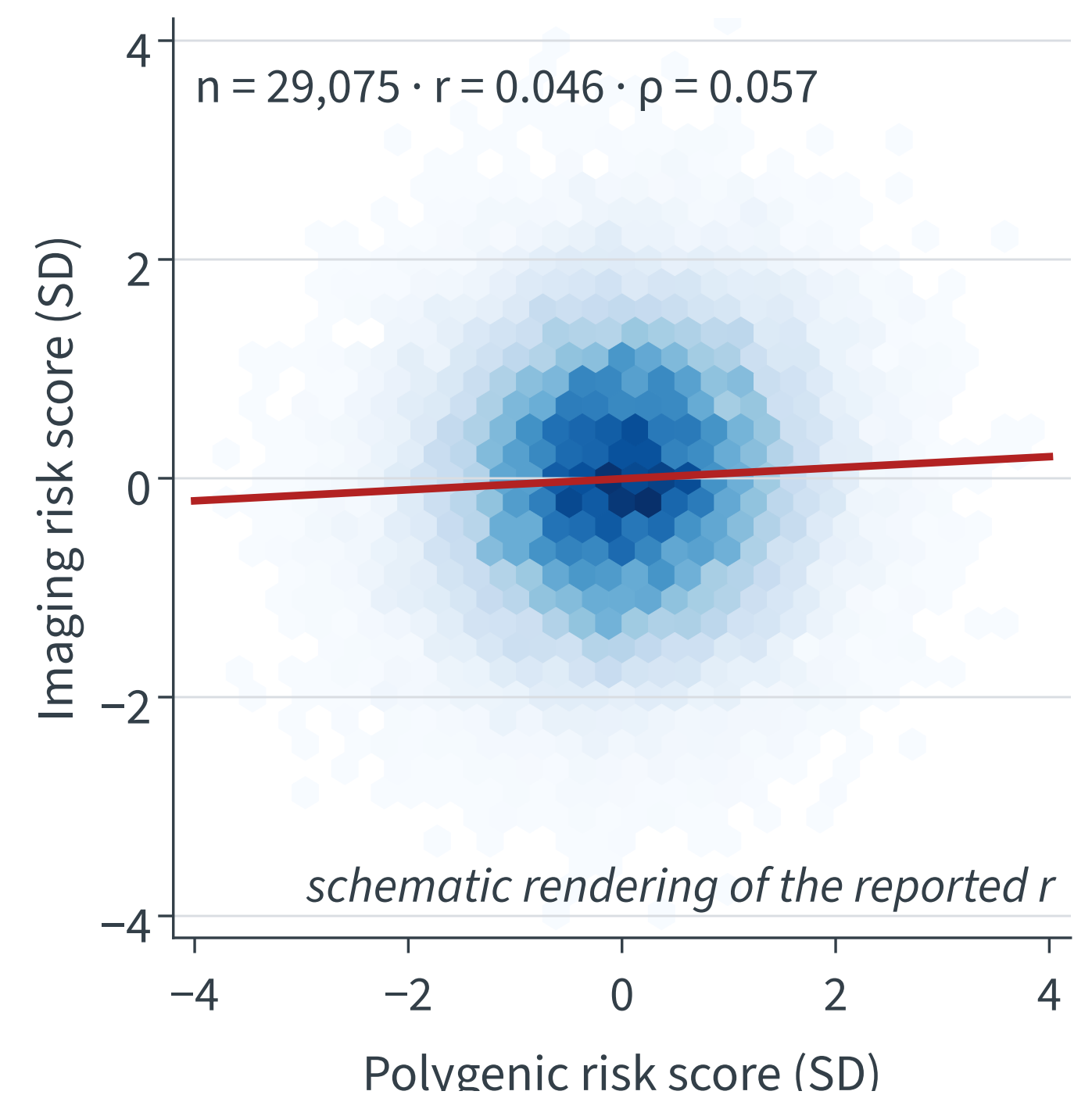
Embeddings encode physiology: CV1 loads on stroke volume, ejection fraction, myocardial mass and wall thickness (Fig. 2c).



Medical pretraining wins: Med3D for the cardiac 3D encoders, RadImageNet for liver, pancreas and DXA (axes from 0.50 / 0.45; Supp. Fig. 3b, 4b).

IMAGING VS GENETICS

Imaging and genetic risk are nearly independent



r ≈ 0.05

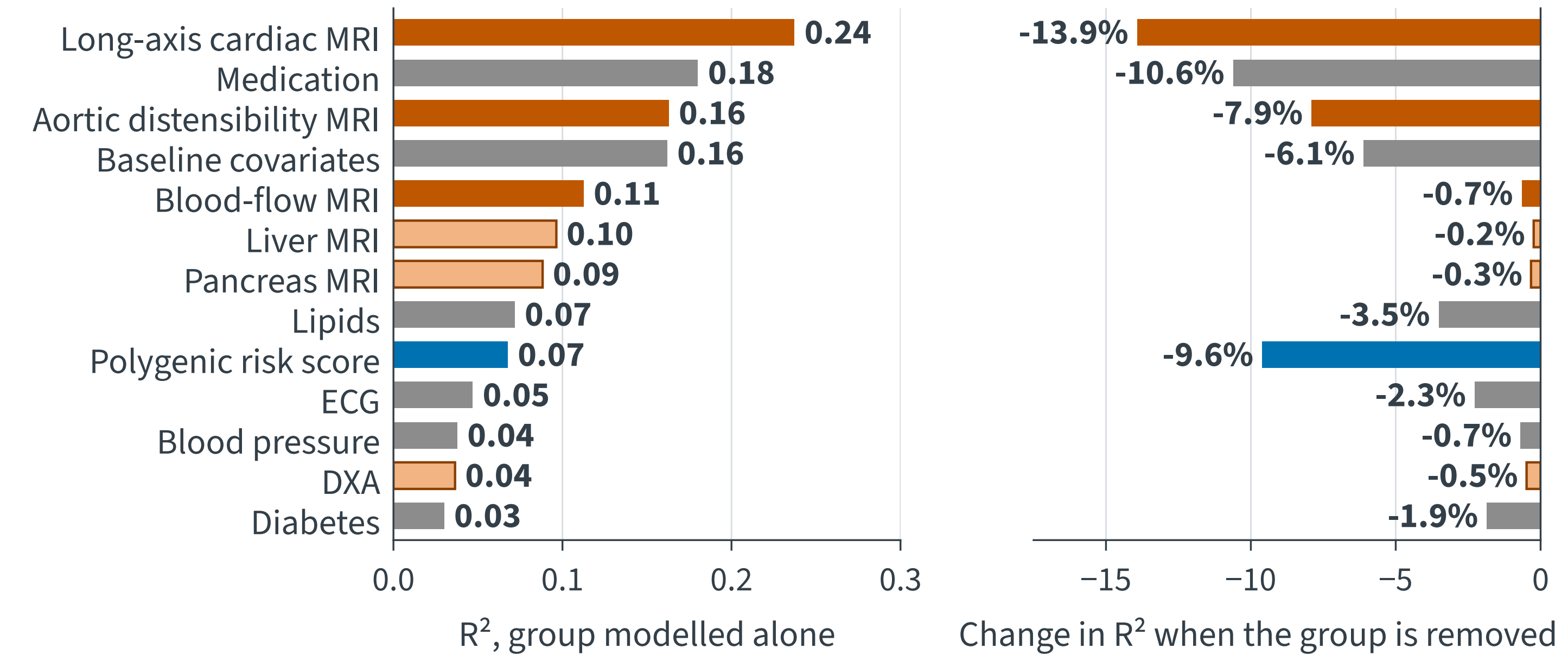
Pearson r, imaging vs polygenic risk score ($\rho = 0.057$); $R^2 < 1\%$ shared — the two overlap very little ($n = 29{,}075$; Fig. 3c–e).

Partial R^2 of imaging risk explained by

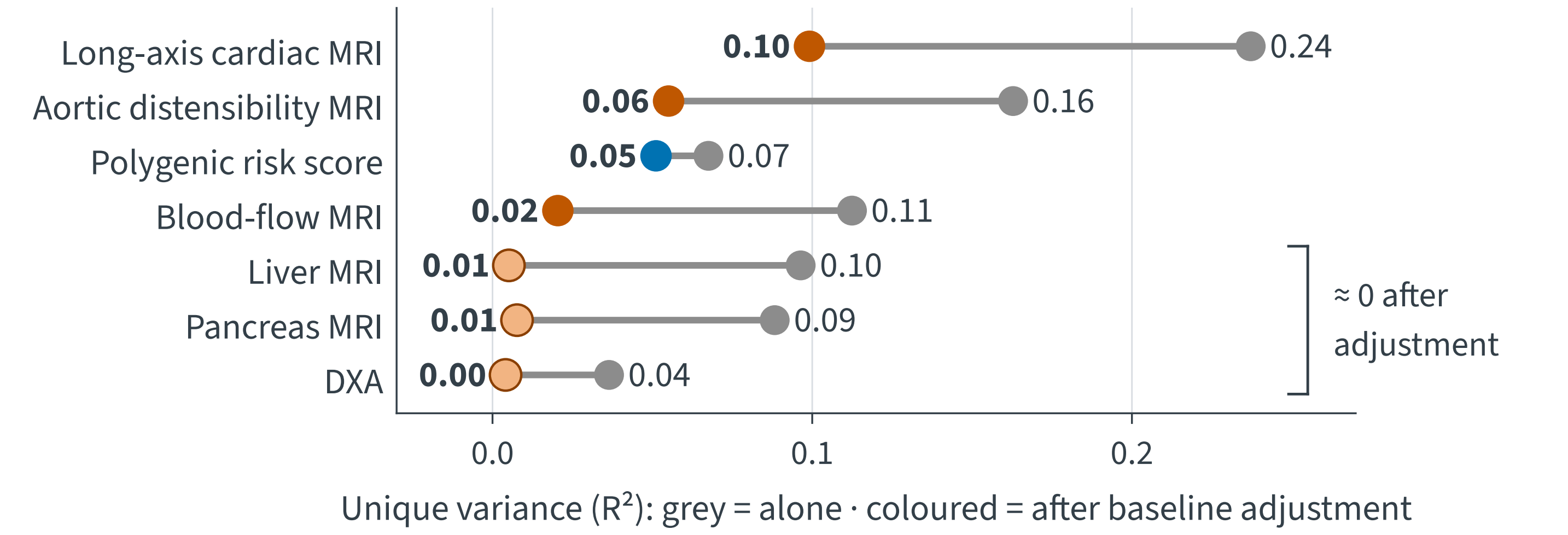
0.15% genetics (PRS)

38.4% demographics

14.4% clinical factors



Cardiac imaging, medication and PRS are each non-redundant. Left: R^2 of each group alone; right: R^2 drop when the group is removed from the full model (Fig. 3a,b).



Only long-axis, aortic distensibility MRI and PRS retain unique variance after clinical adjustment (Fig. 3f).

INTEGRATION & STRATIFICATION

Sequential integration finds very-high-risk people

36.8%

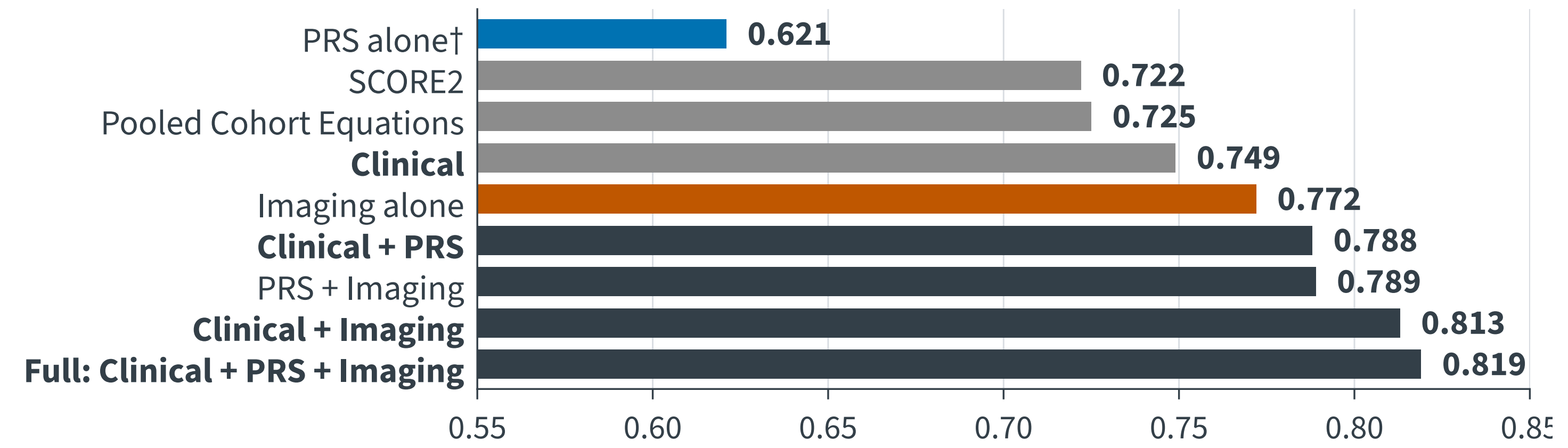
5-yr CAD incidence, top quintile of both imaging and genetic risk within Tier 3; cohort 1.6%

0.634
continuous NRI vs clinical (95% CI 0.534–0.734)

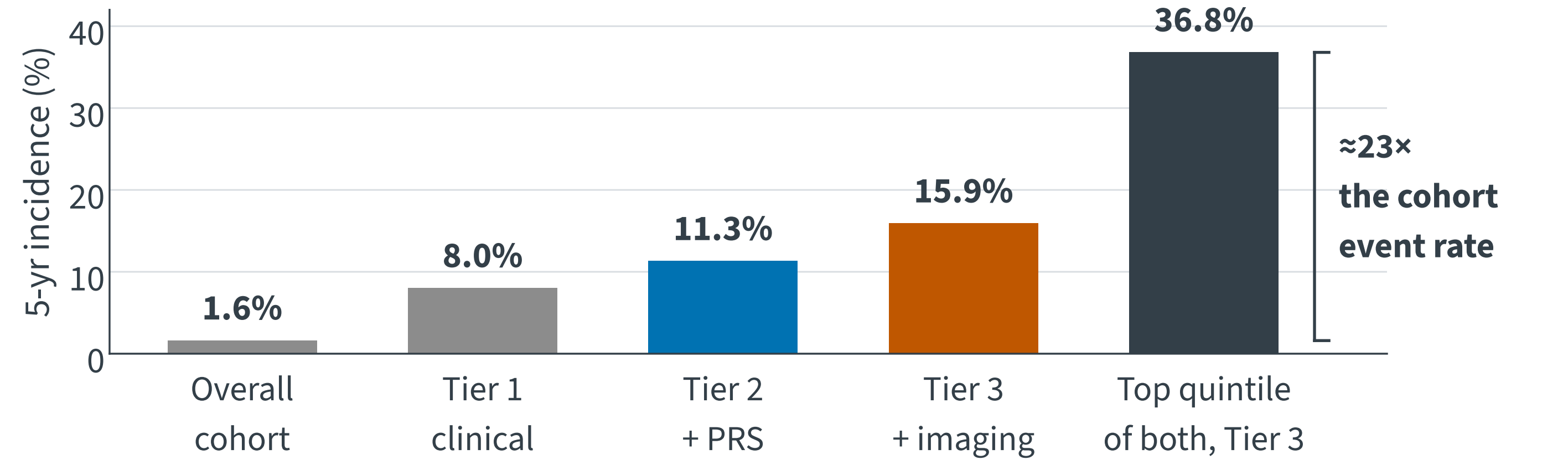
9.5%
prediction accuracy (IPA), full model; clinical 5.5%

86.8 yr
median heart age, Tier 3; cohort 55.3 yr

0.99–1.08
calibration slopes, nested models



Each layer adds discrimination; imaging is among the strongest independent predictors (HR ≈ 2.1 per SD). Held-out C-index; axis from 0.55; bold = nested. HR per SD, full model: imaging ≈ 2.1, PRS ≈ 1.7, demographics ≈ 1.4, lipids ≈ 1.1. †PRS alone = score without age/sex; AUC 0.766 (column 1) includes age, sex, 20 PCs.



Sequential tiers enrich risk far beyond clinical stratification (5-yr incidence, held-out set). Leave-one-centre-out: AUC 0.785–0.822, C-index 0.73–0.76; discrimination is stable across the four UK Biobank imaging centres (site-level robustness, not external validation).

Conclusions

- Imaging and polygenic risk are nearly uncorrelated: complementary axes.
- Long-axis cine and aortic distensibility MRI carry the incremental signal; blood-flow, liver, pancreas and DXA add little once clinical variables are included.
- Clinical → PRS → imaging integration isolates a 36.8% 5-yr-risk subgroup.
- A framework for complementary risk — not yet a clinical screening tool.

Limitations

- No CAC/CTA in UK Biobank: no calcium-score benchmark.
- European-ancestry, complete-case, healthier-than-general cohort; covariates measured at imaging, so part of imaging's gain may reflect temporal proximity.
- Not externally validated; observational — prospective trials needed.